

Istiaq Ahmed Mahim

+880 1727-422285 | mahim.istiakahmed@gmail.com | Dhaka, Bangladesh
LinkedIn | GitHub | Portfolio | Codeforces

OBJECTIVE

12th-semester Computer Science and Engineering student seeking a **Network Engineering / NOC internship**, with hands-on ISP exposure in routing, switching, router and IP configuration, CAT6/RJ45 cabling, and basic network troubleshooting. Experienced with Cisco Packet Tracer and OSPF-based multi-site network design, with research interest in 6G URLLC and dynamic network slicing. Motivated to support reliable, scalable network infrastructure while building deeper expertise in enterprise routing, switching, and network operations.

EDUCATION

International University of Business Agriculture and Technology (IUBAT)	2023 - Present
Bachelor of Science in Computer Science and Engineering	Dhaka, Bangladesh
12th Semester CGPA: 3.56 / 4.00	
Sherpur Government College - HSC	2020 GPA: 5.00 / 5.00
Nabarun Public School - SSC	2018 GPA: 5.00 / 5.00

TECHNICAL SKILLS

Networking: Routing & Switching, OSPF, Static Routing, IPv4 Addressing, Subnetting, VLANs, Trunking, Inter-VLAN Routing, Router Configuration, LAN Setup, RJ45 Crimping, CAT6 Cabling, Basic Network Troubleshooting

Tools & Platforms: Cisco Packet Tracer, Visual Studio Code, Git, GitHub, Arduino IDE

Programming Languages: C, C++, C#, JavaScript

Web & Database: HTML, CSS, Node.js, SQL

Productivity & Media: Microsoft Word, Excel, PowerPoint, Canva, CapCut

HANDS-ON NETWORKING EXPERIENCE

Practical Networking Experience - ISP Environment

- Worked in an ISP environment with practical exposure to **routing, switching, router configuration, and IP configuration**.
- Configured routers and client IP settings, assisted LAN connectivity, and supported basic connectivity troubleshooting.
- Prepared Ethernet connections by crimping **RJ45 connectors on CAT6 cable** and gained hands-on experience with physical network setup.

RESEARCH

Latency-Aware Federated Learning with Dynamic Network Slicing for 6G URLLC under Heterogeneous Constraints	Research Paper
---	----------------

6G • URLLC • Federated Learning • Dynamic Network Slicing • MEC • IoV • Reinforcement Learning

– Developed an experimental framework for latency-aware learning and dynamic resource allocation under heterogeneous edge-network conditions.

– Evaluated centralized neural learning, 9-client FedAvg, asynchronous federated learning, and RL-based allocation; reported **97.30% RL decision accuracy** and a **3.36% simulated delay reduction** for asynchronous FL versus synchronous training.

PROJECTS

TechNova Solutions Network Design Using OSPF Routing	Networking Project
Cisco Packet Tracer • OSPF • VLAN • Router-on-a-Stick • IPv4/Subnetting • Routing & Switching	

- Designed and configured a multi-branch enterprise network connecting **Dhaka, Chattogram, and Sylhet** using OSPF dynamic routing.
- Implemented VLANs, trunking, router subinterfaces, inter-VLAN routing, /24 LANs and /30 WAN links; verified routes and connectivity with Cisco IOS show commands, ping, and traceroute.

Smart Disaster Monitoring Network Using Arduino and Multi-Sensor Edge-Based Risk Detection

Arduino / IoT

Arduino Uno R3 • DHT11 • BMP280 • Rain Sensor • Vibration Sensor • Processing 4.5.6 • CSV

- Built an environmental monitoring system for temperature, humidity, atmospheric pressure, rainfall/wetness, and vibration with real-time risk classification.
- Designed laptop-side visualization and CSV logging for live sensor values, risk status, and disaster-management recommendations.

Military Surveillance Rover

Arduino / Robotics

Arduino Uno • Ultrasonic Sensor • DHT11 • Servo • LCD • IP Camera • C/C++

- Developed a remotely controlled surveillance rover with obstacle detection, environmental sensing, visual alerts, and live IP-camera monitoring.

Green City Planner & Healthy Settlements - NASA

GitHub

Space Apps Challenge 2025

Next.js • React • Node.js • Leaflet • NASA Earth Data • Environmental APIs • jsPDF

- Built a city-level environmental decision-support dashboard that combines Earth-observation and open data for air quality, weather, flood/fire risk, humidity, and planning recommendations.

AgroProcureBD - Smart Agricultural Marketplace

Software Project

& Procurement Platform

React • TypeScript • Node.js • Express.js • MySQL • REST API • JWT

- Contributed to frontend development and database management for a multi-role agricultural marketplace and simplified procurement platform.

AI-Based Expert System for Loan Approval

AI / ML

Decision Support

Python • Random Forest • Gaussian Naive Bayes • Best-First Search • Scikit-learn • Pandas

- Built a hybrid rule-based and machine-learning decision-support system; the Random Forest model achieved **98.01% accuracy** on the project dataset.

Additional Projects: Movie Ticket Management System (ASP.NET Core MVC, SQL Server) • Inventory Management System for GESS Bangladesh (PHP, MySQL) • AI Tool Usage Prediction Among Students (Logistic Regression)

ACHIEVEMENTS & ACTIVITIES

- **NASA International Space Apps Challenge 2025 - Galactic Problem Solver / Participant;** contributed to Team CodexGalaxy and the Green City Planner project.
- **Volunteer, Python Power-Up Session** - BASIS Students' Forum, IUBAT Chapter; recognized with a Certificate of Appreciation.
- **Volunteer, “The Drone Revolution: From Hobby to National Security”** - BASIS Students' Forum, IUBAT Chapter, 27 April 2025.

LANGUAGES, SOFT SKILLS & INTERESTS

Language: English - Fluent in reading, writing, and speaking

Soft Skills: Leadership, Communication, Teamwork

Hobbies & Interests: Swimming, Running, Chess, Football